Analysis of Variance (ANOVA) is a statistical method to compare the means of multiple groups. Multivariate Analysis of Variance (MANOVA) extends ANOVA to multivariate data, comparing mean vectors between groups. In this report, we compare issues with running ANOVA after finding a significant difference between groups with MANOVA. This can dramatically inflate the overall type I error rate. In this report, we showed this by repeatedly running MANOVA on samples from the same trivariate normal distribution for each group. If the result was significant, we then ran univariate ANOVAs for each variable. After running all simulations using a significance level of 0.05, MANOVA had an expected type I error rate of 0.047. The univariate ANOVAs had type I error rates of 0.426, 0.383, and 0.383.